

# WCH11 — Rapid Revision

IAL Chemistry Unit 1: Structure, Bonding and Introduction to Organic Chemistry

Last-minute cram sheets — designed to be scanned, not read

Topic priority ordered by frequency across 22 past papers (Jan 2019 – May 2026)

80 marks · 90 minutes

Section A: 20 multiple-choice (20 marks) · Section B: 60 marks

## TRIAGE — WHAT TO REVIEW, IN WHAT ORDER

| Time left  | Cover these                                                                                         | Why                                                                  |
|------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 30 min     | Sheets 2, 6, 8 — $PV = nRT$ , ionisation energy, molecular shapes                                   | 22/22, 21/22 and 19/22 papers. Highest marks per minute.             |
| 2 hours    | Add sheets 1, 3, 5, 11 — moles, formulae and atom economy, atomic structure, electrophilic addition | All in 14–22 papers. "Carbocation" appeared 70x in the mark schemes. |
| Half a day | Add sheets 4, 7, 9, 10, 12                                                                          | Bonding, structure, organic naming, radicals, polymers.              |
| Always     | Sheet 13 (phrase triggers)                                                                          | 4.3 "explain" commands per paper.                                    |

### THREE THINGS SPECIFIC TO THIS PAPER

- $PV = nRT$  was in all 22 papers.** Units are the whole difficulty — sheet 2.
  - Do NOT revise titration (0/22 papers), intermolecular forces (1/22) or electronegativity (1/22).** They belong to other units.
  - Atom economy beats percentage yield 17 papers to 4.** Most students prepare the reverse.
- Timing:** 22 min on Section A · 58 min on Section B · 10 min to check. Never leave an MCQ blank.

## SHEET 1 — MOLES & REACTING MASSES (22/22 papers)

$n = m / M$  moles = mass (g) ÷ molar mass ·  $n = c \times V$  V in  $\text{dm}^3$  ·  $n = V / 24$  gas at RTP, V in  $\text{dm}^3$  (or +24000 for  $\text{cm}^3$ ) ·  **$N = n \times 6.02 \times 10^{23}$**

**$1 \text{ dm}^3 = 1000 \text{ cm}^3$**  — divide  $\text{cm}^3$  by 1000 before using  $n = cV$ . Most common slip in the topic.

**Reacting mass method:** **1.** balanced equation · **2.** convert the given quantity to **moles** · **3.** use the **molar ratio** · **4.** convert back to what is asked.

**Limiting reagent** (when two reactant amounts are given): divide moles of each by its coefficient — **the smaller value is limiting**, and all later working uses it.

## SHEET 2 — $PV = nRT$ (22/22 papers — every single one)

### THE UNITS ARE THE WHOLE DIFFICULTY

**P** in **Pa** —  $\text{kPa} \times 1000$  · **V** in  **$\text{m}^3$**  —  $\text{cm}^3 \div 10^6$ ,  $\text{dm}^3 \div 1000$  · **n** in **mol** · **R** = 8.31 (given) · **T** in **K** —  $^\circ\text{C} + 273$

Rearrangements:  **$n = PV/RT$**   **$V = nRT/P$**   **$T = PV/nR$**   **$M = mRT/PV$**

**$M = mRT/PV$**  is how the paper asks for a relative molecular mass from gas data — it has appeared in exactly that form repeatedly. **Write the unit conversions as separate lines before substituting.** Doing them in your head is where the marks go.

## SHEET 3 — FORMULAE, YIELD & ATOM ECONOMY (16/22 and 17/22)

### Empirical formula — five steps

**1.** % or mass of each element · **2.** ÷ by its  **$A_r$**  · **3.** ÷ all by the **smallest** · **4.** multiply to whole numbers ( $.5 \rightarrow \times 2$ ,  $.33/.67 \rightarrow \times 3$ ) · **5.** molecular formula = ( $M_r$  ÷ empirical formula mass) × empirical formula.

**Hydrated salt, find x:** mass of water lost → moles; mass of anhydrous salt → moles; take the **ratio water : salt**.

**atom economy** = ( $M_r$  of desired product ÷ total  $M_r$  of all products) × 100

**percentage yield** = (actual ÷ theoretical) × 100

**The distinction tested:** atom economy is a property of the **equation**, fixed before you run anything. Percentage yield is a property of **that run**. An addition reaction has **100% atom economy** (one product) and can still give a poor yield.

**Why high atom economy matters:** less waste → lower disposal cost and less environmental harm; more efficient use of **finite** raw materials. Two points = the mark.

## SHEET 4 — EQUATIONS & SIGNIFICANT FIGURES

- Ionic equations:** full equation → split aqueous ionic substances into ions → cancel **spectator ions** → check **atoms AND total charge** balance.
- State symbols** (s) (l) (g) (aq) are a mark in their own right whenever asked for.
- Significant figures:** specified in **17/22** papers. Give what is asked, or match the **least precise data**. Keep full precision through intermediate steps; **round only at the end**.

## SHEET 5 — ATOMIC STRUCTURE, ISOTOPES & MASS SPEC (18/22 and 10/22)

| Particle | Relative mass       | Relative charge |
|----------|---------------------|-----------------|
| Proton   | 1                   | +1              |
| Neutron  | 1                   | 0               |
| Electron | 1/1836 (negligible) | -1              |

**Isotopes** — same **protons**, different **neutrons**. **Identical chemical properties** (same electron configuration); physical properties differ slightly (mass).

**$A_r = \Sigma(\text{isotopic mass} \times \% \text{ abundance}) \div 100$** . If given as peak heights, divide by the **total of the heights**, not by 100.

**Definition:** the **weighted mean mass of an atom** relative to **1/12 the mass of an atom of carbon-12**.

**Mass spectrometry, in order:** **ionisation** → **acceleration** → **deflection** (by mass-to-charge ratio) → **detection**.

**m/z** = ion mass (charge is almost always +1) · highest m/z peak = **molecular ion** =  $M_r$ .

**Cl** gives **3:1** ( $^{35}\text{Cl}$  :  $^{37}\text{Cl}$ ) · **Br** gives **1:1** · **Cl<sub>2</sub>** gives peaks at **70, 72, 74 in 9:6:1** — a favourite question.

## SHEET 6 — ELECTRON CONFIGURATION & IONISATION ENERGY (21/22 papers — most examined explanation topic)

**Filling order:** 1s 2s 2p 3s 3p **4s 3d** 4p. **4s fills before 3d but EMPTIES FIRST** — ions lose 4s electrons first.

**Two exceptions:** Cr = [Ar] 3d<sup>5</sup>4s<sup>1</sup>, Cu = [Ar] 3d<sup>10</sup>4s<sup>1</sup> (half-filled / full d sub-shell is more stable). s holds 2, p holds 6, d holds 10. Fill p orbitals **singly first**, parallel spins, then pair with opposite arrows.

### DEFINITION — EXACT WORDING

Energy to remove **one mole of electrons** from **one mole of gaseous atoms** to form **one mole of gaseous 1+ ions**.  $X(g) \rightarrow X^+(g) + e^-$

2nd IE:  $X^+(g) \rightarrow X^{2+}(g) + e^-$ . **The (g) is compulsory** and is the most frequently lost mark.

| Factor                | Effect on ionisation energy                                   |
|-----------------------|---------------------------------------------------------------|
| <b>Nuclear charge</b> | More protons → stronger attraction → <b>higher</b>            |
| <b>Atomic radius</b>  | Outer electron further out → weaker attraction → <b>lower</b> |
| <b>Shielding</b>      | More inner shells screen the nucleus → <b>lower</b>           |

**Down a group — decreases:** larger radius and more shielding, which **outweigh** the increased nuclear charge.

**Across a period — increases:** nuclear charge rises, shielding stays similar, radius decreases.

**Dip at Group 3** (Al < Mg): outer electron is in a **p orbital, higher in energy**, removed more easily.

**Dip at Group 6** (S < P): outer p orbital has **two paired electrons which repel**, so less energy needed.

**Successive IEs:** a **large jump** = next electron from a **shell closer to the nucleus with less shielding**. Electrons removed **before** the jump = number in the outer shell = the **group**.

## SHEET 7 — BONDING & DOT-AND-CROSS (13/22 papers)

| Bond                   | Definition — learn the wording                                                                | Examples                                                                                       |
|------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Ionic</b>           | <b>Electrostatic attraction between oppositely charged ions</b> , formed by electron transfer | Metal + non-metal                                                                              |
| <b>Covalent</b>        | A <b>shared pair of electrons</b> attracted to both nuclei                                    | Non-metal + non-metal                                                                          |
| <b>Dative covalent</b> | <b>Both electrons from the same atom</b> — one with a <b>lone pair</b>                        | $\text{NH}_4^+$ , $\text{H}_3\text{O}^+$ , $\text{NH}_3\text{BF}_3$ , $\text{Al}_2\text{Cl}_6$ |
| <b>Metallic</b>        | <b>Electrostatic attraction between positive metal ions and delocalised electrons</b>         | Metals                                                                                         |

A dative bond is drawn as an **arrow from donor to acceptor**, and once formed is **identical in strength and length** to any covalent bond — often a mark.

### DOT-AND-CROSS CONVENTIONS

**Different symbols** for electrons from different atoms, labelled · **ionic**: square brackets round each ion, **charge outside**, metal ion's outer shell **empty** · **outer shell only** unless told otherwise · **covalent**: show **all lone pairs** · check the total electron count against the formula and charge.

## SHEET 8 — SHAPES & BOND ANGLES (19/22 papers · "lone pair" ×115 in the mark schemes)

### THE PRINCIPLE, WORD FOR WORD

Electron pairs **repel** and arrange themselves **as far apart as possible**, giving minimum repulsion. **Lone pairs repel more strongly than bonding pairs**, so each lone pair **reduces the bond angle by about 2.5°**.

| Bond pairs | Lone pairs | Shape                | Angle               | Example                         |
|------------|------------|----------------------|---------------------|---------------------------------|
| 2          | 0          | Linear               | <b>180°</b>         | $\text{CO}_2$ , $\text{BeCl}_2$ |
| 3          | 0          | Trigonal planar      | <b>120°</b>         | $\text{BF}_3$ , $\text{SO}_3$   |
| 4          | 0          | Tetrahedral          | <b>109.5°</b>       | $\text{CH}_4$ , $\text{NH}_4^+$ |
| 3          | 1          | Trigonal pyramidal   | <b>107°</b>         | $\text{NH}_3$ , $\text{PCl}_3$  |
| 2          | 2          | Bent / V-shaped      | <b>104.5°</b>       | $\text{H}_2\text{O}$            |
| 5          | 0          | Trigonal bipyramidal | <b>120° and 90°</b> | $\text{PCl}_5$                  |
| 6          | 0          | Octahedral           | <b>90°</b>          | $\text{SF}_6$                   |

Draw in **3D with wedges and dashed lines**, label the **angle**, show **lone pairs**. A flat tetrahedral drawing loses the mark.

## SHEET 9 — STRUCTURE & PROPERTIES (12/22 giant covalent, 9/22 metallic)

| Structure                                        | Properties, and the reason                                                                                                                                                                                    |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Giant ionic</b>                               | High m.p. — strong attraction between oppositely charged ions throughout the lattice. Conducts <b>molten or dissolved only</b> , because <b>ions are only then free to move</b> . Brittle.                    |
| <b>Simple molecular</b>                          | Low m.p. — only the <b>weak intermolecular forces</b> are overcome, <b>not</b> the covalent bonds. Does not conduct.                                                                                          |
| <b>Giant covalent</b><br>diamond, $\text{SiO}_2$ | Very high m.p., hard, non-conducting — continuous network of strong covalent bonds, no free charged particles.                                                                                                |
| <b>Graphite</b>                                  | <b>Conducts</b> — each C bonds to only <b>three</b> others, leaving <b>one delocalised electron per atom</b> . <b>Soft and slippery</b> — layers slide over one another.                                      |
| <b>Giant metallic</b>                            | Conducts in all states (mobile delocalised electrons). <b>Malleable</b> — layers of ions slide without breaking the bonding. m.p. rises with <b>ionic charge</b> and <b>number of delocalised electrons</b> . |

### THE SENTENCE THAT LOSES MOST MARKS

"The covalent bonds break when it melts." For **simple molecular** substances, melting overcomes the **intermolecular forces only**. Write **"intermolecular forces are overcome"**.

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## SHEET 10 — ORGANIC NAMING & ISOMERISM (10/22 structural isomers)

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**Naming:** longest chain containing the functional group → number from the end giving **lowest locants** → substituents **alphabetically** → di/tri/tetra for repeats → commas between numbers, hyphens between numbers and letters.

**Formula types:** empirical (simplest ratio) · molecular (actual) · general ( $C_nH_{2n+2}$  alkanes,  $C_nH_{2n}$  alkenes) · structural (condensed) · displayed (every bond) · skeletal (lines, H omitted).

**Structural isomerism:** chain (branching) · position (group moved) · functional group (different group).

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### E/Z ISOMERISM

Requires **restricted rotation about C=C** AND **two different groups on each carbon** of the double bond.

Priority by **atomic number**. Both priority groups on the **same side** → **Z**; **opposite sides** → **E**. *Cis/trans* only when each carbon carries one H.

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## SHEET 11 — MECHANISMS (19/22 papers · "carbocation" ×70 in mark schemes)

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### Free radical substitution — alkanes, UV light (10/22)

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**Initiation**  $Cl_2 \rightarrow 2Cl\cdot$  (homolytic fission, by UV)

**Propagation**  $Cl\cdot + CH_4 \rightarrow \cdot CH_3 + HCl$  then  $\cdot CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\cdot$

**Termination** any two radicals combining

**Homolytic fission** = **one electron to each atom** → two radicals. **Show the dot**. Propagation **regenerates** the radical — that's why it's a chain.

**Why a poor synthesis:** gives a **mixture** — further substitution to  $CH_2Cl_2$ ,  $CHCl_3$ ,  $CCl_4$ , and at different positions.

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### Electrophilic addition — alkenes (14/22)

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$C=C$  is a **sigma** + **pi bond**; the pi bond is **high electron density**, attacked by **electrophiles** (electron pair acceptors).

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#### EVERY FEATURE IS A MARK

**1.** Curly arrow from the **C=C** to the **6+ atom** · **2.** curly arrow from the **breaking bond** to the **more electronegative atom** · **3.** **carbocation** with the charge on the **correct carbon** · **4.** curly arrow from the **lone pair or negative charge** to the positive carbon · **5.** product.

Curly arrows show **a pair of electrons** and start from a **bond, lone pair or negative charge** — never an atom symbol. With  $Br_2$ , say the dipole is **induced**.

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| Reagent / conditions            | Product        | Note                                            |
|---------------------------------|----------------|-------------------------------------------------|
| $Br_2$ , room temperature       | Dibromoalkane  | <b>Orange → colourless</b> — the test for $C=C$ |
| $HBr$ / $HCl$                   | Halogenoalkane | Markovnikov applies                             |
| Conc. $H_2SO_4$ , then water    | Alcohol        | Two steps                                       |
| Steam, $H_3PO_4$ , high T and p | Alcohol        | Industrial ethanol                              |
| $H_2$ , Ni, ~150 °C             | Alkane         | Hydrogenation                                   |

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### MARKOVNIKOV — THE REASON IS THE A\* MARK

Major product forms via the **more stable carbocation**. Stability: **tertiary** > **secondary** > **primary**, because alkyl groups are **electron-donating and reduce the positive charge** on the carbon. Stating the order without that reason scores half.

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## SHEET 12 — CRUDE OIL, COMBUSTION & POLYMERS (4/22, 5/22, 17/22)

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**Fractional distillation** separates by **boiling temperature** (longer chains → stronger forces → condense lower). **Cracking**: long alkanes → shorter alkanes + alkenes. *Thermal* = high T and p, favours alkenes; *catalytic* = **zeolite**, lower T, favours branched alkanes and aromatics for motor fuel.

**Incomplete combustion** → **carbon monoxide** (toxic — binds haemoglobin in place of oxygen) and **particulates**. **Other pollutants**: **SO<sub>2</sub>** from sulfur impurities → acid rain; **NO<sub>x</sub>**, formed from N<sub>2</sub> and O<sub>2</sub> at high engine temperatures → acid rain and photochemical smog; unburnt hydrocarbons. **Catalytic converter**: 2CO + 2NO → 2CO<sub>2</sub> + N<sub>2</sub>.

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### ADDITION POLYMERS (17/22)

Double bond **opens**, monomers join, **no other product** → **100% atom economy**.

**Repeat unit**: open the double bond, two carbons joined by a single bond, keep all substituents, **extended bonds through the brackets, n** outside. Missing either = lost mark. Monomer from polymer: take one repeat unit, put the double bond back.

**Disposal**: landfill — wastes space, **non-biodegradable** because the **C-C backbone is unreactive** · incineration — releases energy but gives CO<sub>2</sub>, and toxic HCl from PVC · recycling — conserves finite crude oil but needs sorting · feedstock recycling — back to monomers.

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## SHEET 13 — PHRASE TRIGGERS

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| If the question is about... | ...the phrase they want is                                                                              |
|-----------------------------|---------------------------------------------------------------------------------------------------------|
| First ionisation energy     | <b>One mole of electrons</b> from <b>one mole of gaseous atoms</b> → <b>one mole of gaseous 1+ ions</b> |
| IE down a group             | <b>Larger radius + more shielding</b> , which <b>outweigh</b> the greater nuclear charge                |
| IE across a period          | <b>Nuclear charge increases</b> , shielding similar, <b>radius decreases</b>                            |
| Dip at Group 3              | Outer electron in a <b>p orbital</b> , <b>higher in energy</b>                                          |
| Dip at Group 6              | <b>Paired electrons in a p orbital repel</b>                                                            |
| Big jump in successive IEs  | Electron from a <b>shell closer to the nucleus with less shielding</b>                                  |
| Molecular shape             | Pairs <b>repel</b> , go <b>as far apart as possible</b> ; <b>lone pairs repel more</b> , ~2.5° each     |
| Ionic bonding               | <b>Electrostatic attraction between oppositely charged ions</b>                                         |
| Metallic bonding            | <b>Electrostatic attraction between positive ions and delocalised electrons</b>                         |
| Melting a molecular solid   | <b>Intermolecular forces are overcome</b> — never "covalent bonds break"                                |
| Graphite conducting         | Each C bonds to <b>three</b> others → <b>one delocalised electron per atom</b>                          |
| Ionic solid not conducting  | <b>Ions not free to move</b> in the solid                                                               |
| Isotopes                    | Same <b>protons</b> , different <b>neutrons</b> ; <b>identical chemical properties</b>                  |
| Homolytic fission           | <b>One electron to each atom</b> , forming two <b>radicals</b>                                          |
| Carbocation stability       | <b>Alkyl groups are electron-donating</b> → <b>reduce the positive charge</b>                           |
| Test for C=C                | Bromine water <b>decolourised</b> , <b>orange to colourless</b>                                         |
| Atom economy benefit        | <b>Less waste</b> , lower disposal cost; efficient use of <b>finite</b> raw materials                   |
| Polymers persisting         | <b>C-C backbone is unreactive</b> → <b>non-biodegradable</b>                                            |

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## SHEET 14 — 60-SECOND SELF-TEST

Cover the right column. Hesitate on one, go back to that sheet.

| Question                                             | Answer                                                                                   |
|------------------------------------------------------|------------------------------------------------------------------------------------------|
| Units for $P$ , $V$ and $T$ in $PV = nRT$ ?          | Pa, m <sup>3</sup> , K                                                                   |
| Rearrange $PV = nRT$ for $M_r$ .                     | $M = mRT/PV$                                                                             |
| Convert 250 cm <sup>3</sup> to m <sup>3</sup> .      | $2.5 \times 10^{-4} \text{ m}^3$ ( $\div 10^6$ )                                         |
| How do you find the limiting reagent?                | Divide moles of each by its coefficient; smaller value is limiting                       |
| Atom economy formula?                                | $(M_r \text{ of desired product} \div \text{total } M_r \text{ of products}) \times 100$ |
| Atom economy of any addition reaction?               | 100% — there is only one product                                                         |
| Relative atomic mass definition?                     | Weighted mean mass of an atom relative to 1/12 the mass of a carbon-12 atom              |
| Four stages of mass spectrometry?                    | Ionisation, acceleration, deflection, detection                                          |
| Cl <sub>2</sub> molecular ion peaks and ratio?       | 70, 72, 74 in 9:6:1                                                                      |
| Electron configuration of Cr?                        | [Ar] 3d <sup>5</sup> 4s <sup>1</sup>                                                     |
| Which electrons does a transition metal lose first?  | 4s before 3d                                                                             |
| First ionisation energy equation?                    | $X(g) \rightarrow X^+(g) + e^-$                                                          |
| Three factors affecting ionisation energy?           | Nuclear charge, atomic radius, shielding                                                 |
| Why the dip at Group 6?                              | Paired electrons in a p orbital repel                                                    |
| What does a large jump in successive IEs show?       | The next electron is from a shell closer to the nucleus                                  |
| Define a dative covalent bond.                       | Both electrons come from the same atom                                                   |
| Bond angle in NH <sub>3</sub> ? In H <sub>2</sub> O? | 107°; 104.5°                                                                             |
| Bond angles in PCl <sub>5</sub> ?                    | 120° and 90°                                                                             |
| How much does one lone pair reduce a bond angle?     | About 2.5°                                                                               |
| Why does graphite conduct?                           | Each carbon bonds to only three others, leaving one delocalised electron per atom        |
| What is overcome when a molecular solid melts?       | Intermolecular forces — not covalent bonds                                               |
| Two conditions for E/Z isomerism?                    | Restricted rotation about C=C; two different groups on each carbon                       |
| Z or E — priority groups on the same side?           | Z                                                                                        |
| Conditions for free radical substitution?            | Ultraviolet light                                                                        |
| Why is it a poor synthetic method?                   | It gives a mixture of products                                                           |
| What is the intermediate in electrophilic addition?  | A carbocation                                                                            |
| Carbocation stability order, and why?                | Tertiary > secondary > primary; alkyl groups are electron-donating                       |
| Test for an alkene?                                  | Bromine water decolourised, orange to colourless                                         |
| Two things a repeat unit must show?                  | Extended bonds through the brackets, and $n$                                             |
| Why are addition polymers non-biodegradable?         | The C-C backbone is unreactive                                                           |

### WALKING INTO THE EXAM

Convert every unit before substituting, especially in  $PV = nRT$ . Show every calculation line. Check the significant figures asked for. Put **(g)** in every ionisation energy equation. Draw shapes in 3D with the angle labelled. Start curly arrows from electron pairs, never atoms. Answer every multiple-choice question — there is no penalty for a wrong one.